

Advanced OS Monitor Project(OVERVIEW):

Project Description

This project is a real-time system monitoring dashboard built using Python. It works like a simplified version of Windows Task Manager.

It tracks:

CPU usage ,RAM usage ,Disk usage ,Internet speed ,Running processes ,CPU scheduling algorithms (FCFS & Round Robin) ,User activity (typing + idle time).

How It Works

The program continuously collects system data using:

`psutil` → CPU, RAM, Disk, Process info

`speedtest` → Internet speed

`tkinter` → GUI interface

`matplotlib` → CPU usage graph

`pynput` → keyboard activity tracking

All data is updated every 1 second in real time.

Features

- ✓ Real-time CPU / RAM / DISK monitoring
- ✓ Colored progress bars (visual indicators)
- ✓ Live CPU usage graph
- ✓ Top running processes display
- ✓ Internet speed monitor
- ✓ CPU scheduling simulation:

FCFS (First Come First Serve)

Round Robin (RR)

- ✓ System Health Score (0–100)
- ✓ CPU overload warning system
- ✓ User typing + idle tracking

Output

 ADVANCED OS MONITOR

SYSTEM HEALTH: 74/100 ↘

🔍 INTERNET → ↓ 28 Mbps ↑ 6 Mbps

🔍 FCFS → Wait: [0,5,8] Avg: 4.33

🔍 RR → Wait: [6,4,9] Avg: 6.33

CPU : 52%

RAM : 72%

DISK : 41%

🔍 chrome.exe using 25% CPU

🔍 CPU LOAD

🖱️ Typing: 132

🔍 Idle: 2.8 sec

- CPU → Blue progress bar
- RAM → Purple progress bar
- DISK → Yellow progress bar
- CPU graph → Live line chart

Purpose of Project

- This project demonstrates:
- Operating System concepts
- CPU scheduling algorithms
- Real-time system monitoring
- GUI development in Python
- Data visualization

This project helps understand OS resource management and scheduling algorithms in real-time environment.

Output SS below:

🔧 ADVANCED OS MONITOR

SYSTEM TIME: 11:18:38 04-05-2026

SYSTEM HEALTH: 43/100 ⚡

CPU : 21.6%

RAM : 64.6%

DISK : 86.3%

🖨 Typing: 760
🕒 Idle: 0.19 sec

● CPU OK ⚠ System Idle Process using 931.8% CPU

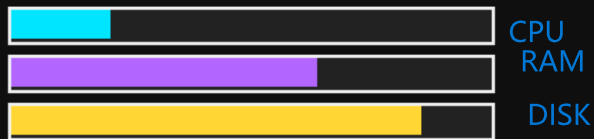
🌐 INTERNET → ↓ 81.21 Mbps ↑ 54.33 Mbps

📦 FCFS → Wait: [0, 5, 8] Avg: 4.33

📦 RR (Q=2) → Wait: [7, 6, 8] Avg: 7.00

TOP PROCESSES:

System Idle Pro	931.8%
msedgewebview2.	100.0%
python.exe	36.9%
svchost.exe	22.6%
PhoneExperience	16.3%



CPU Usage History

